

Chapter 01: Why Deep Learning Exists

Significance of DL

DL is one of the most influential technologies of the 21st century.

A bit history

"Let's build a model with 100 million parameters" (DL) was a crazy talk even during 2010, when we had data, CPUs, Classical ML— all fine.

A few years later from 2010, models began outperforming humans in image recognition. Then they learn speech. Then language. Then strategy games. Then programming. Then art. Then scientific discovery. We can easily guess that someone invented a new algorithm. But that would be a wrong guess!

The core idea of DL—artificial neural networks—had existed since the 1940s and 1950s. Then, the revolution happened after 2010 because several missing pieces finally came together.

What is AI about?

Artificial Intelligence (AI) is the broad field of creating machines capable of performing tasks that normally require human intelligence.

Tasks like:

- Understanding language
- Recognizing faces
- Driving cars
- Playing chess
- Diagnosing diseases

- Translating languages
- Writing software

Early AI relied heavily on manually written rules. This approach quickly becomes impossible for complex tasks (imagine, writing rules for recognizing every cat).

What ML offers

ML offered to allow computers to learn from data— structured data— structured in columns and rows— engineered/manually-extracted features and targets as columns, and observations as rows.

Now, the model discovers the rules itself. Better than Rule-Based AI Model.

ML replaced AI? No. ML became a valuable part of AI, by extending what AI can accomplish.

Where Classical ML Begins to Struggle

ML (then ML, now Classical ML) usually requires feature engineering. Humans had to manually create useful features.

Imagine teaching a computer to recognize cats.

From a Raw Image, a human immediately notices:

- ears
- whiskers
- eyes
- fur
- body shape

But computers see only numbers. The digital representation of the Raw Image is an array of numbers. In ML, the computer does not see even this array of numbers.

Feature engineering is one of the biggest ideas in classical ML. From Raw Image, humans design features, and these features/structured data is fed to the algorithm. In ML, the human, still, does much of the intelligence.

The computer sees [125, 144, 233, ...], an array of million numbers, for a cat, but this array is not even the array of numbers that is the digital representation of the raw image. The computer/algorithm never truly "sees" the cat. It only sees the features we designed. This approach has worked remarkably well for many years but not for problems like **Recognizing a cat from an image**.

Suppose we want to recognize:

- every dog in an image
- every bird
- every face
- every handwritten digit
- every spoken word

How many handcrafted features would we need?

- Thousands
- Millions
- Different features for every problem

The complexity explodes.

Humans become the bottleneck.

What DL offers

DL says, "Instead of asking humans to design features, let the computer learn the features." Yes, no more Feature Engineering by humans.

For **Recognizing a Cat in an Image**,

ML:

Image

-> Human manually crafts/engineers Features (*edges, curves, textures, shapes*)

-> ML Model

-> Prediction

DL:

Image

-> Neural Network

The DL Neural Network learns *edges, curves, textures, shapes object parts, complete objects*, on its own from the Raw Image.

-> Prediction

Representation Learning

The neural network learns useful internal representations of the data, on its own, from raw data, unlike ML which needs structured data, with no explicit programming which is like ML, yes. This trait is called Representation Learning.

The Neural Network, during training, discovers useful internal representations of the data, and builds a hierarchy of learned representations.

An Example Hierarchy of Learned Representations

Pixels

-> Edges

-> Corners

-> Simple Shapes

-> Eyes

-> Face

-> Person

The layers in the hierarchy become increasingly abstract from detail.

As we stack more layers, the network can learn increasingly abstract representations.

This *depth* gives deep learning its expressive power.

What is 'Deep'?

Deep Learning = Learning with deep (multi-layer) neural networks.

Why did the revolution happen after 2010?

The core idea of DL—artificial neural networks—had existed since the 1940s and 1950s. Then, the revolution happened after 2010 because several missing pieces finally came together.

The missing pieces

- **Massive Datasets:** Massive Datasets, with millions and billions of training examples, are needed for DL. Such datasets became available around 2012, through the internet and large-scale data collection.

- **Powerful Hardware:** GPU with enormous Parallel Computing Power is needed for efficient DL. GPUs reached this level of development around 2012.
- **Better Algorithms:** Meanwhile, there were improvements even in training methods, optimization algorithms, and neural network architectures.
- **Better Softwares:** Frameworks like PyTorch and TensorFlow dramatically reduced the complexity of building and training neural networks.
- **More Research:** As results improved, investment and research accelerated, leading to rapid innovation and a virtuous cycle of progress.

Scope of Modern DL

Computer Vision, NLP, Speech, Science, Robotics, Recommendation Systems etc.

DL replaced ML?

Just like ML didn't replace AI, rather ML became a valuable part of AI, by extending what AI can accomplish, DL too simply extended what ML and AI can accomplish, thus becoming a valuable part of ML and AI.

Deep learning is not the best choice for every problem.

Classical machine learning is often preferable when:

- you have a small dataset,
- interpretability is crucial,
- computational resources are limited, and
- the problem is relatively simple.

A good engineer chooses the right tool for the task, not the most fashionable one.